

Total No. of Questions : 4]

SEAT No. :

PF2

[Total No. of Pages : 2

APR-26/FE/Insem-2

F.E. (Insem)

ENGINEERING CHEMISTRY

(2019 Pattern) (Semester - II) (107009) (Credit System)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q. No. 1 or Q. No. 2 and Q. No. 3 or Q. No. 4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicates full marks.
- 4) Use of logarithmic tables, slide rule, mollier charts, pocket electronic calculator is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Explain demineralization method with principle, figure, ion exchange reaction and regeneration for softening of impure water. **[5]**

b) Explain the difference between scales and sludges. **[4]**

c) What is EDTA? Draw M-EDTA complex, structure and give chemical reactions involved. **[3]**

d) 50 ml water sample required 12.1 ml of N/50 HCl upto phenolphthalein end point and total 18.5 ml of the same acid upto methylorange end point. Calculate type and amount of alkalinity present in water sample. **[3]**

OR

Q2) a) Define scales. Explain any three causes and preventive measure of scales in boiler. **[5]**

b) What is hardness of water? Explain the difference between temporary hardness and permanent hardness. **[4]**

c) Explain reverse osmosis process with figure and advantages. **[3]**

d) A zeolite bed exhausted by softening 4000 litres of water requires 10 litres of 15% NaCl solution for regeneration. Calculate the hardness of water sample. **[3]**

P.T.O.

- Q3)** a) Explain three stages of pH meter titration between strong acid with strong base with titration curve and reaction. [5]
- b) Define reference electrode. Give the construction of calomel electrode with labelled diagram. [4]
- c) Define : [3]
- i) Ion selective electrode
 - ii) Molar conductance
 - iii) Cell constant
- d) Give the composition of membrane of ISE for the determination of H^+ , F^- and Cl^- . [3]

OR

- Q4)** a) Explain the three stages of conductometric titration of weak acid and weak base with titration curve and reaction. [5]
- b) Which electrodes are used for measurement of pH. Explain the standardization process of pH meter. [4]
- c) Define : [3]
- i) Buffer
 - ii) Equivalent conductance
 - iii) Specific conductance
- d) Explain the construction of conductivity cell with labelled diagram. [3]

